SWABRI KANENJE Generative AI Engineer (Foundation Models & Agentic Systems) | IoT & Embedded Systems Engineer

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Profile Summary

A multidisciplinary engineer working at the intersection of Generative AI, agentic systems, and IoT. I specialize in building **foundation-model–powered applications** with strong alignment to modern ML application roles: prompt engineering, evaluation frameworks, retrieval-augmented generation, safe deployment patterns, and scalable backend integration.

My background in **IoT systems engineering** and **embedded programming** gives me an edge in bridging physical devices with AI-driven cloud intelligence—designing systems where sensors, edge devices, and foundation models work together to produce real-time insights.

I combine hands-on embedded experience (Go, Rust, C/C++) with emerging GenAI application skills (RAG pipelines, model evaluation, LLM agents, and CI/CD for AI systems), enabling me to deliver intelligent, secure, production-aligned AI features.

Core Skills

CATEGORY	SKILLS & TOOLS
GENERATIVE AI APPLICATION	Foundation Model Evaluation & Selection, RAG, Prompt Engineering, LLM Agents, Guardrails & Safety, Model Monitoring, CI/CD for AI Apps
PROGRAMMING	C/C++ (Embedded), Python (Data Handling & Analytics), Go (Concurrent Device Services), Rust (Low-Level Performance)
DATA & DB	SQL (MySQL, MariaDB), SQLite, NoSQL (JSON), File I/O
MLOPS & DEVOPS	Git, GitHub Actions (CI/CD), Docker, AWS, Agile Methodologies, Scalability Optimization
FOUNDATIONS	Electrical Circuit Theory, Control Systems, Digital Signal Processing, Data Structures & Algorithms

Work Experience

SOFTWARE DEVELOPER / DATA SCIENCE FOCUS | Jan 2024 – Present - Zone 01 Kisumu

Full-Stack Developer Training Program

Data Processing Optimization: Designed and implemented scalable backend services in Go for efficient data ingestion and preprocessing, improving raw data processing efficiency by 25% through optimized memory management and concurrent programming.

- Pipeline Automation (MLOps Foundation): Integrated CI/CD pipelines with GitHub Actions to automate the build, test, and deployment phases, reducing the time required to deploy new features from hours to under 10 minutes.
- Code Quality & Collaboration: Conducted regular code reviews and contributed to reusable component libraries, accelerating feature development cycles by 40% and enforcing disciplined software quality standards.
- Front-End Experience: Built responsive, cross-browser compatible frontends with HTML, CSS, and JavaScript, demonstrating an understanding of the full application stack from data visualization to user interaction.
- Agile Practice: Collaborated in peer-to-peer projects, applying Agile and DevOps practices to manage tasks and ensure the timely delivery of complex project milestones

Education: Diploma Electrical Engineering – Dedan Kimathi University Of Technology